

Attack 1: S3 Bucket Public Exposure

Timestamp: 11:14:39

Detection Evidence: AWS Config Rule s3-bucket-public-read-prohibited transitioned to NON_COMPLIANT

Log Proof: CloudTrail Event ID 2a60ac3d-2f2b-4cb4-891a-b7c923975a79 shows PutBucketPublicAccessBlock event on resource test-bucket-simulation-222234573

MITRE ATT&CK Mapping: T1530 (Data from Cloud Storage Object)

Root Cause: Block public access was disabled and the credentials file was made public via ACL, leading to credential exposure

Automated Action: Flagged by AWS Config; no active Lambda blocking for this specific misconfiguration.

Attack 2: IAM Credential Misuse from External IP

Timestamp: 11:59:33 (from alarm state change)

Detection Evidence: IAMMisuse-Alarm triggered to ALARM state

Log Proof: CT logs confirm recon API calls like ListUsers, DescribeInstances, ListBuckets from the external ip called 42.106.176.183.

MITRE ATT&CK Mapping: T1078 (valid Accounts)

Root Cause: The attacker successfully authenticated using the leaked keys from the attack 1 to perform initial account reconnaissance.

Automated Action: lambda fn security auto remediate successfully parsed the alarm and extracted the src ip and injected a DENY rule into the N/W ACL and SNS alert was dispatched.

Attack 3: External Port Scan on EC2

Timestamp: 12:36:42 (from alarm state change)

Detection Evidence: CloudWatch PortScan-Alarm triggered to ALARM state

Log Proof: VPC Flow Logs (/aws/vpc/flowlogs) show high-volume connection tracking from external attacker IP 42.106.176.183 probing multiple destination ports

MITRE ATT&CK Mapping: T1046 (Network Service Discovery)

Root Cause: A public-facing EC2 instance exposed an active network footprint to the open internet, allowing external enumeration

Automated Action: Alarm notification delivered via SNS to email

Attack 4: IAM Privilege Escalation

Timestamp: 12:48:27 (from alarm state change)

Detection Evidence: CloudWatch PrivEsc-Alarm triggered to ALARM state

Log Proof: CloudTrail logs show AttachUserPolicy event for user dev-test involving policy ARN for AdministratorAccess

MITRE ATT&CK Mapping: T1098 (Account Manipulation)

Root Cause: A user with policy-attachment permissions successfully assigned themselves full administrative rights

Automated Action: Lambda function triggered and Alarm notification delivered via SNS to email

Attack 5: API Enumeration (Admin Recon)

Timestamp: 12:58:01 (from alarm state change)

Detection Evidence: CloudWatch APIBurst-Alarm triggered to ALARM state

Log Proof : CloudTrail logs confirm a sudden burst of 27 API calls executed by dev-test

MITRE ATT&CK Mapping: T1580 (Cloud Infrastructure Discovery)

Root Cause: The privilege escalation in Attack 4 granted the attacker the permissions required to perform widespread account enumeration

Automated Action: Lambda function triggered and Alarm notification delivered via SNS to email

Attack 6: Internal Lateral Movement & Port Scan

Timestamp: 13:44:47 (from alarm state change)

Detection Evidence: CloudWatch LateralMovement-Alarm transitioned to ALARM state

Log Proof : VPC Flow Logs (/aws/vpc/flowlogs) show internal traffic originating from the public subnet (10.0.1.206) targeting the private subnet (10.0.2.240)

MITRE ATT&CK Mapping: T1021 (Remote Services), T1046 (Network Service Discovery)

Root Cause: The public web server was compromised and used as a bridgehead to reach an internal network segment that was not adequately isolated from the public-facing tier

Automated Action: The LateralMovement-Alarm triggered the EventBridge/Lambda pipeline, and an SNS alert was dispatched